Generalizable and Per-Scene NeRF Architectures under Sparse-View Conditions: A Systematic Benchmarking Methodology

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Abstract. *This work proposes a methodology for systematically evaluating the trade-offs between computational performance and reconstruction fidelity of Neural Radiance Field (NeRF) architectures under sparse-view conditions. Four representative methods are evaluated using visual fidelity metrics, including Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS), together with computational performance metrics, including training convergence time, Video Random Access Memory (VRAM) footprint, and inference latency. The experimental results reveal a clear trade-off between reconstruction fidelity and computational efficiency. Per-Scene models consistently outperform Generalizable models in visual quality, achieving up to 27.11 dB PSNR, 0.875 SSIM, and 0.171 LPIPS, while also providing significantly higher inference throughput and lower memory consumption. In contrast, generalizable models offer greater flexibility for acquisition-constrained scenarios but exhibit limited reconstruction fidelity and unstable optimization under severe sparse-view conditions. The proposed methodology and the resulting quantitative analyses establish a standardized and reproducible framework for benchmarking heterogeneous neural rendering architectures, enabling systematic model selection according to computational and acquisition constraints.*

1. Introduction

Neural Radiance Fields (NeRFs) have established themselves as the state of the art for novel view synthesis, enabling high-fidelity reconstruction of three-dimensional scenes from multi-view observations across a broad range of computer vision and computer graphics applications [Kerbl et al. 2023, Tancik et al. 2023, Adobe 2025]. Nevertheless, most neural rendering techniques require substantial computational resources and

densely sampled multi-view training data, which restricts their applicability to resource-constrained scenarios such as mobile scanning, virtual try-on systems, and sparse-view reconstruction.

Recent advances in neural rendering have focused on improving the computational efficiency and generalization capability of NeRF-based methods. Representative approaches include Instant Neural Graphics Primitives (Instant-NGP), Nerfacto, PixelNeRF, and Generalizable NeRF Transformer (GNT) [Müller et al. 2022, Tancik et al. 2023, Yu et al. 2021, Wang et al. 2023]. These methods explore different design strategies, including efficient spatial encoding mechanisms [Müller et al. 2022], hybrid architectures that integrate Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) [Schwarz et al. 2021, Wu et al. 2021, Wang et al. 2023], and model pre-computation (baking) into compact representations with reduced memory requirements [Hedman et al. 2021]. Although these developments have significantly improved different aspects of neural rendering, their computational characteristics and reconstruction performance have generally been evaluated independently or under different experimental settings.

Consequently, a systematic and reproducible comparison of generalizable and per-scene neural rendering architectures under standardized sparse-view protocols remains unavailable. Existing studies have primarily investigated NeRF models in limited-data scenarios [Niemeyer et al. 2022, Heo et al. 2023], whereas benchmarking frameworks such as NerfBaselines [Kulhanek and Sattler 2025] focus predominantly on traditional and per-scene methods. This hinders a comprehensive characterization of the trade-offs between computational efficiency and reconstruction fidelity across different neural rendering paradigms.

To address this limitation, a systematic evaluation framework for neural rendering architectures under sparse-view conditions is proposed here. Reconstruction fidelity was quantified using three widely adopted image quality metrics, namely Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [Wang et al. 2004], and Learned Perceptual Image Patch Similarity (LPIPS) [Zhang et al. 2018]. Computational performance was quantified through complementary system-level metrics, including training convergence time, inference latency, and Video Random Access Memory (VRAM) consumption, enabling a quantitative characterization of the trade-offs between computational cost and reconstruction quality. Thus, the main contributions of this work are summarized as follows:

- A unified evaluation protocol was defined for generalizable and per-scene NeRF architectures, ensuring reproducibility through standardized data partitions, controlled execution environments, and consistent evaluation metrics across three sparse-view data regimes.
- Integration wrappers extending the Nerfstudio framework with PixelNeRF and GNT were implemented to provide a unified experimental infrastructure for standardized benchmarking of heterogeneous neural rendering architectures. Consequently, these modules were publicy released, enabling the deployment of these models across broader applications.

2. Theoretical Background

2.1. Neural Radiance Fields and Volumetric Rendering

The foundational architecture of NeRFs utilizes Multilayer Perceptrons (MLPs) to approximate a continuous 5D scene representation [Mildenhall et al. 2020]. This function takes a spatial position $\mathbf{x} = (x, y, z)$ and a viewing direction $\mathbf{d} = (\theta, \phi)$ as inputs, outputting the volumetric density $\sigma(\mathbf{x})$ and associated color $\mathbf{c} = (r, g, b)$:

$$\gamma(\mathbf{x}, \mathbf{d}) \to \mathbf{v}, \quad F'_\theta(\mathbf{v}) \to (\sigma, \mathbf{c}), \quad \sigma \in \mathbb{R}_{\geq 0},\ \mathbf{c} \in [0, 1]^3$$

where θ represents the neural network weights, F'_θ is the original MLP, and γ is a positional encoder mapping inputs into a higher-dimensional feature vector $\mathbf{v}$.

To synthesize novel views, the expected color of a pixel is computed via volumetric rendering. For a ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$ with depth bounds (t_n, t_f), the resulting color is:

$$\hat{C}(\mathbf{r}) = \int_{t_n}^{t_f} T(t)\sigma(\mathbf{r}(t))\mathbf{c}(\mathbf{r}(t), \mathbf{d})\, dt$$

where $T(t) = \exp\left(-\int_{t_n}^{t} \sigma(\mathbf{r}(s))\, ds\right)$ denotes the accumulated transmittance. In practice, this integral is numerically approximated via alpha compositing over discrete samples along each ray.

2.2. Architectural Optimizations and Generalization

Although the original NeRF achieves high-fidelity results, its reliance on large MLPs imposes computational bottlenecks. To accelerate optimization, hybrid representations were introduced to replace Fourier positional encoding with hash encoding. Instant-NGP proposes a multiscale hash table that maps spatial coordinates to feature vectors, heavily reducing optimization time [Müller et al. 2022]. Nerfacto builds upon this mechanism by combining multiresolution hash encoding with proposal sampling, pose refinement, spherical harmonics, and appearance embeddings to balance visual quality and rendering speed [Tancik et al. 2023].

These per-scene methods, however, fundamentally rely on dense multi-view observations. In practical scenarios, such as mobile scanning, virtual try-on, and medical imaging (for instance, histology and holotomography), multi-view acquisition is often physically limited [Laurino et al. 2023]. Under these sparse-view constraints, lacking camera pose redundancy, per-scene models suffer geometric degradation.

To synthesize novel views from sparse inputs, few-shot generalizable NeRFs incorporate structural priors learned from large datasets via dedicated feature extractors. PixelNeRF applies a pre-trained Residual ResNet to extract spatial features from 2D projections, conditioning the radiance MLP [Yu et al. 2021]. GNT replaces the CNN with a vision transformer for global feature extraction, and substitutes the MLP with a cross-attention transformer to aggregate multi-view information [Wang et al. 2023].

To maximize visual fidelity, generalizable models frequently employ fine-tuning (FT). Instead of purely zero-shot (ZS) inference, FT updates the network weights using

the sparse views of the novel scene as direct photometric supervision. This rapid optimization reduces the domain gap, though it carries the risk of representational collapse if overfitted.

Given the fundamental architectural differences between per-scene optimizations and generalizable priors, selecting the optimal model for sparse-view deployments remains non-trivial. Consequently, a standardized evaluation framework is essential to systematically benchmark these trade-offs, which forms the core methodology of this work.

3. Materials and Methods

This section presents the proposed methodology for systematically evaluating the trade-offs between computational performance and reconstruction fidelity of NeRF models under sparse-view conditions. Figure 1 illustrates the proposed evaluation pipeline, which consists of three main stages: (i) dataset selection and sparse-view sampling; (ii) configuration and training of the selected neural rendering architectures according to their respective learning paradigms; and (iii) quantitative evaluation using standardized metrics to jointly characterize computational efficiency and reconstruction fidelity.

3.1. Dataset & Splits

The experiments were conducted on the Local Light Field Fusion (LLFF) dataset, which comprises 8 real-world scenes captured with a forward-facing moving camera. Camera poses and depth bounds were obtained through COLMAP, following standard dataset procedures [Mildenhall et al. 2019]. Furthermore, images were evaluated using downscale of 8 resolution (504×378).

The complete data splitting protocol follows the few-shot evaluation configuration established by Niemeyer et al. [Niemeyer et al. 2022]. Specifically, the testing split was generated by reserving images with indices that are multiples of 8 (llffhold=8), while the training views for the sparse regimes were strictly limited to $N \in \{3, 6, 10\}$. The selection of these training views was performed using two distinct strategies: (i) uniform, where views are sampled via linear interpolation over the dataset indices, distributing them evenly along the camera trajectory (a strategy that does not generalize to other datasets, since LLFF indices encode camera trajectory order); and (ii) random, where views are randomly selected using a fixed seed (42) to ensure reproducibility. Comparing these two strategies assesses the models sensitivity to the spatial distribution of the input views, as demonstrated in Table 1. Given the empirical advantage of uniform sampling (+0.54 dB PSNR on the fern scene), this strategy was adopted for all subsequent experiments.

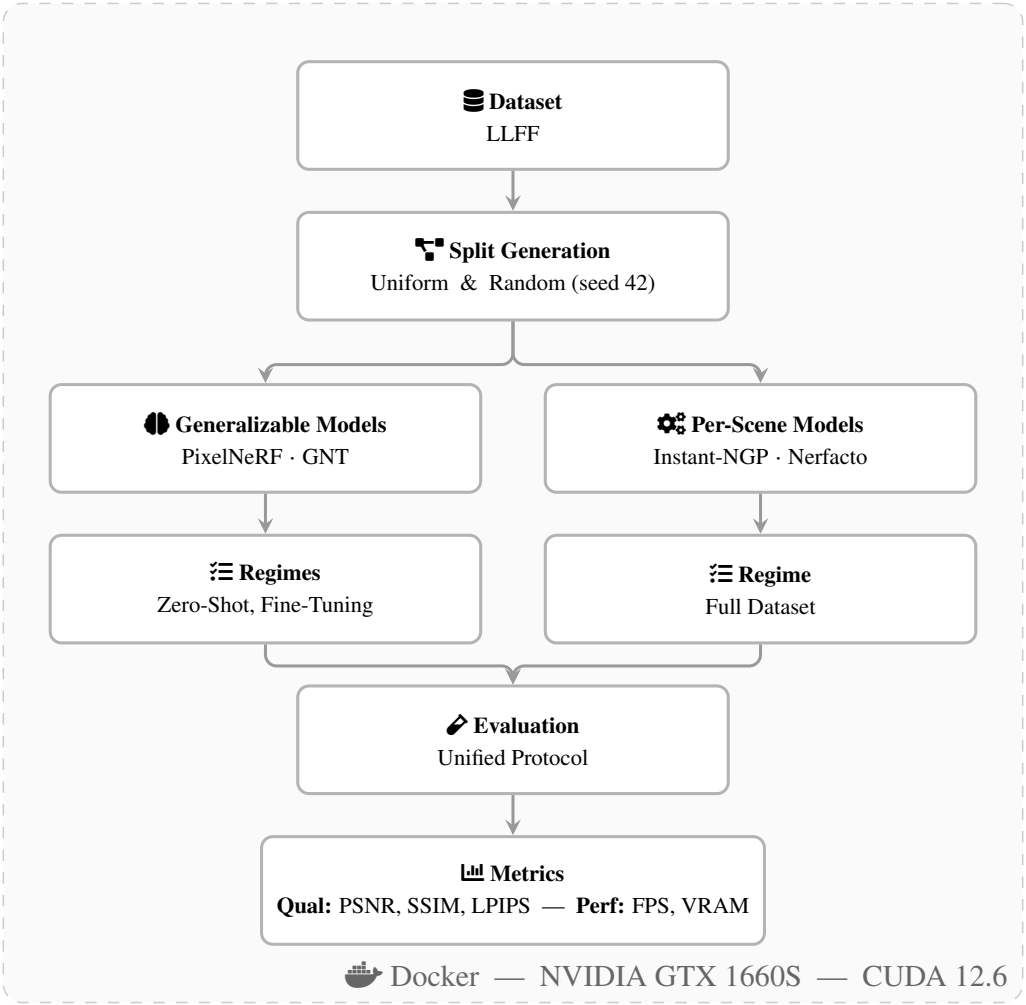


Figure 1. Overview of the proposed methodology. The pipeline comprises dataset preparation, sparse-view sampling, model configuration and training under distinct learning paradigms, followed by a unified evaluation protocol that systematically characterizes the trade-offs between computational performance and reconstruction fidelity.

Table 1. Impact of the training view selection strategy (uniform versus random) on average visual quality. Evaluated under the fine-tuning regime with $N = 10$ views on the fern scene.

Model	Split	PSNR ↑	SSIM ↑	LPIPS ↓	ΔPSNR
PixelNeRF	Uniform	6.98	0.047	0.846	**+0.54**
	Random	6.44	0.042	0.840	
GNT	Uniform	10.34	0.179	0.738	+0.15
	Random	10.19	0.184	0.742	
Instant-NGP	Uniform	**13.57**	**0.231**	**0.627**	+0.06
	Random	13.51	0.229	0.630	
Nerfacto	Uniform	12.04	0.227	0.809	+0.11
	Random	11.93	0.226	0.865	

Specific pipeline configurations were strictly enforced to guarantee stability and fair comparison. Notably, camera pose refinement was disabled for Nerfacto to keep all

models on equal footing, as this feature is not natively supported by the other evaluated architectures. To minimize memory footprint and accelerate fine-tuning, mixed-precision training was enabled across all models. For hash-based models (Instant-NGP and Nerfacto), the multiscale hash table capacity was constrained to 2^{16}, with backgrounds set to white. Structural hyperparameters for GNT were locked to 48 coarse and importance samples per ray, a transformer depth of 8, and a network width of 64, following its original configuration [Wang et al. 2023].

3.2. Models & Training Regimes

The evaluation encompasses distinct architectural paradigms: generalizable models (PixelNeRF [Yu et al. 2021] and GNT [Wang et al. 2023]) and highly efficient per-scene baselines (Instant-NGP [Müller et al. 2022] and Nerfacto [Tancik et al. 2023]).

Models were evaluated across three regimes: (i) ZS: instant inference using pre-trained weights without local optimization; (ii) FT: fine-tuning pre-trained generalizable models using sparse input views ($N \in \{3, 6, 10\}$); and (iii) Per-Scene: optimizing baseline models from scratch. For direct comparison, per-scene models were trained strictly under the same sparse regimes ($N \in \{3, 6, 10\}$). During FT, the core feature extraction networks were kept frozen, updating only the terminal rendering layers to prevent catastrophic forgetting and overfitting observed in preliminary tests.

Table 2. Impact of N training views on PSNR (dB) on the room scene.

Model	Regime	N=3 ↑	N=6 ↑	N=10 ↑
PixelNeRF	ZS	7.35	7.15	7.29
	FT	7.24	7.26	7.13
GNT	ZS	**12.96**	**12.85**	**12.85**
	FT	11.78	**12.85**	11.58

Table 3. PSNR (dB) under varying learning rates (η). Ablation conducted on the fern scene under the fine-tuning regime with $N = 10$ views.

Model	Regime	Learning Rate (η)	PSNR ↑
PixelNeRF	FT	10^{-4}	11.19
	FT	5×10^{-4}	12.23
	FT	10^{-3}	9.42
	ZS	N/A	6.64
GNT	FT	10^{-4}	**12.96**
	FT	5×10^{-4}	**12.96**
	FT	10^{-3}	12.85
	ZS	N/A	10.73

To ensure equitable evaluation, all models were unified within the Nerfstudio framework [Tancik et al. 2023]. While Instant-NGP and Nerfacto are natively supported, custom integration wrappers for PixelNeRF and GNT were developed and publicly released. These wrappers map their default architectures into Nerfstudio's modular components without altering native inference logic, leveraging tiny-cuda-nn [Müller 2021] for GPU acceleration.

For consistency, all models used Adam ($\eta = 10^{-3}$, zero weight decay) with 512 training and 128 evaluation rays per batch. Fine-tuning ran for 2500 iterations, and per-scene optimizations for 30000. Although per-architecture tuning of η could yield higher peak metrics (e.g., PixelNeRF improved by 2.81 dB with $\eta = 5 \times 10^{-4}$, Table 3), a uniform $\eta = 10^{-3}$ was enforced to isolate architecture as the primary variable, sacrificing absolute per-model performance in favor of a controlled comparative baseline.

The complete wrapper integration code is publicly available at `https://github.com/Felipe-gsilva/Few-Shot-NeRFs/`. Additionally, the modular wrappers can be downloaded directly as packages from PyPI and piwheels `https://www.piwheels.org/project/nerfstudio-gnt/` and `https://www.piwheels.org/project/nerfstudio-pixelnerf/`), while the reproducible experiment scripts and environment configurations are hosted at `https://github.com/Felipe-gsilva/nerf-ann-paper/`.

3.3. Evaluation & Metrics

The evaluation framework assesses three complementary dimensions: visual fidelity, computational efficiency, and memory consumption. Final metrics were computed exclusively on the held-out test set (section 3.1), ensuring no overlap with the training views. To quantify visual quality, this project used three standard novel view synthesis metrics: (i) PSNR, to measure pixel-wise fidelity; (ii) SSIM [Wang et al. 2004], to evaluate structural coherence; and (iii) LPIPS [Zhang et al. 2018]. Following the protocol established by Niemeyer et al. [Niemeyer et al. 2022], LPIPS was computed using Visual Geometry Group (VGG) features to accurately quantify perceptual distance. Operational efficiency was measured through the inference rendering rate in frames per second (fps) across the test cameras, as did Kulhanek et al. [Kulhanek and Sattler 2025]. For memory evaluation, both the static parameter footprint of the models and the dynamic VRAM allocation peak logged by PyTorch during runtime were analyzed. To guarantee environmental isolation and reproducibility, all experiments were executed within Nvidia Docker containers, preventing external processes from adding noise to the performance metrics. The hardware setup consisted of an NVIDIA GeForce GTX 1660 Super (6GB VRAM) running CUDA 12.6.

4. Results and Discussion

This section presents the consolidated quantitative, computational and qualitative analyses across the evaluated LLFF scenes. Table 4 summarizes the global averages for visual fidelity, temporal efficiency, and memory footprint. The results reveal critical trade-offs between generalization, optimization stability, and hardware constraints.

Table 4. Consolidated quantitative results across the LLFF dataset. Metrics represent the global average. Training time reflects the fine-tuning phase for fine-tuning and full convergence for Per-Scene models. VRAM Peak denotes dynamic memory allocation during inference.

Model	Regime	PSNR ↑	SSIM ↑	LPIPS ↓	Train / FT Time ↓	Inference (FPS) ↑	VRAM Peak (GB) ↓
PixelNeRF	ZS	6.62	0.093	0.814	∼22 s	0.004	1.93
	FT	6.38	0.071	0.801	4 min 23 s	0.006	1.97
GNT	ZS	10.73	0.300	0.760	**∼9 s**	0.014	1.94
	FT	10.42	0.336	0.724	7 min 19 s	0.015	1.93
Instant-NGP	N=10	14.06	0.442	0.498	31 min 51 s	0.199	**1.09**
Nerfacto	N=10	**27.11**	**0.875**	**0.171**	8 min 57 s	**0.388**	1.43

4.1. Visual Fidelity

The per-scene architectures demonstrated vastly superior visual reconstruction capabilities. Nerfacto established the performance upper bound, achieving a global average of 27.11 dB PSNR and 0.875 SSIM (Table 4). In contrast, the generalizable models struggled significantly with the sparse constraints of the LLFF dataset.

PixelNeRF produced degenerate reconstructions across all LLFF scenes, consistent with its architectural dependency on dense multi-view overlap for accurate 2D-to-3D feature projection, a condition fundamentally incompatible with the $N \leq 10$ regime evaluated here. The purely local 2D feature projection via the frozen CNN encoder failed to triangulate depth accurately under extreme sparsity, resulting in severe isotropic blurring and a complete failure to reconstruct the underlying geometry. GNT, conversely, achieved structurally superior reconstructions relative to PixelNeRF within the same constrained optimization budget. Although not able to converge to the final scene's minor details, the PSNR values ranged from 11.58 to 12.96, as observed in Table 2. Interestingly, GNT's fine-tuning performance on the room scene degrades from 12.85 dB at $N = 6$ down to 11.58 dB at $N = 10$. It is hypothesized that the fixed $\eta = 10^{-3}$, combined with a denser but still sparse view set, accelerates overfitting of the rendering layers, producing a local performance valley consistent with the representational collapse regime described in Section 2. The inability to fully capture high-frequency, in-the-wild textures highlights the limitations of keeping the feature extraction network frozen during adaptation, forcing the model to rely on generalized priors that ultimately struggle to resolve out-of-distribution geometric complexities.

Furthermore, the stability of these visual results across different scenes is illustrated in Figure 2. The boxplot highlights the significant performance variance inherent to each paradigm. Generalizable models (PixelNeRF and GNT) exhibit a highly constrained PSNR distribution, reflecting a consistent inability to capture high-frequency details regardless of the scene geometry. This variance reflects per-scene optimization sensitivity to geometric complexity; notably, even the worst-case per-scene PSNR exceeds the ceiling of all generalizable configurations. Instant-NGP achieved 19.33 dB in its worst scenario, whereas PixelNeRF's absolute peak was bounded at 12.23 dB under any view budget.

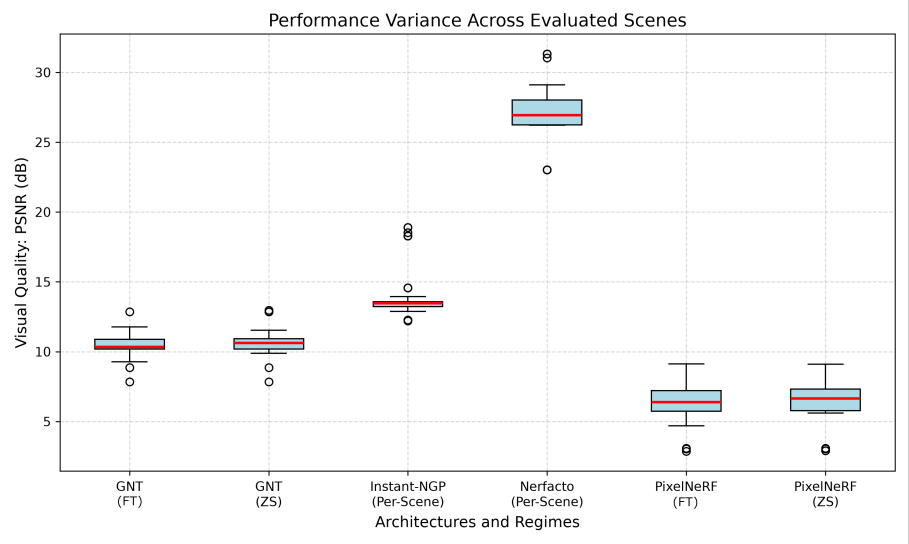


Figure 2. Visual performance variance (PSNR) across all evaluated LLFF scenes.

4.2. Computational Trade-offs

From a systemic viability perspective, a clear dichotomy emerges between inference latency and training overhead. While generalizable models bypass extensive optimization (requiring only seconds for ZS inference or $\sim$7 minutes for FT), their volumetric feature aggregation mechanisms result in catastrophic inference bottlenecks, yielding rendering speeds below 0.02 FPS.

Conversely, the explicit multiscale hash representations of the per-scene models proved vastly more efficient during runtime. Despite requiring lengthier convergence periods (from $\sim$9 minutes for Nerfacto up to $\sim$32 minutes for Instant-NGP), they achieve inference rates ranging from $13\times$ (Instant-NGP vs. GNT) to $97\times$ (Nerfacto vs. PixelNeRF) faster than the generalizable baselines. Furthermore, the hash-based models exhibit a highly optimized dynamic memory footprint. While Instant-NGP operates at just 1.10 GB of VRAM during inference on the fern scene (Table 5), the global average footprint across all scenes scales slightly to 1.43 GB for Nerfacto (Table 4) due to structural complexity variance between captures. Nonetheless, these metrics solidify per-scene methods as the most viable path for hardware-constrained deployments despite the initial training cost.

4.3. Regime Trade-off Synthesis

The architectural trade-offs evaluated establish a rigid Pareto frontier between data availability and computational overhead, as visually summarized by the multidimensional radar chart in Figure 3 . When multi-view data is abundant but inference latency and VRAM are strictly bounded, the per-scene paradigm (specifically Nerfacto) dominates the design space, delivering state-of-art fidelity with sub-second rendering. Conversely, under view scarcity ($N \leq 10$), the structural priors of generalizable models become indispensable to prevent total geometric collapse, yet they exact a severe penalty: inference speeds plummet to impractical levels (≤ 0.02 FPS) and visual quality plateaus prematurely, as the frozen feature extractor cannot adapt its priors to scene-specific geometric complexities (a manifestation of the representational bottleneck described in Section 2).

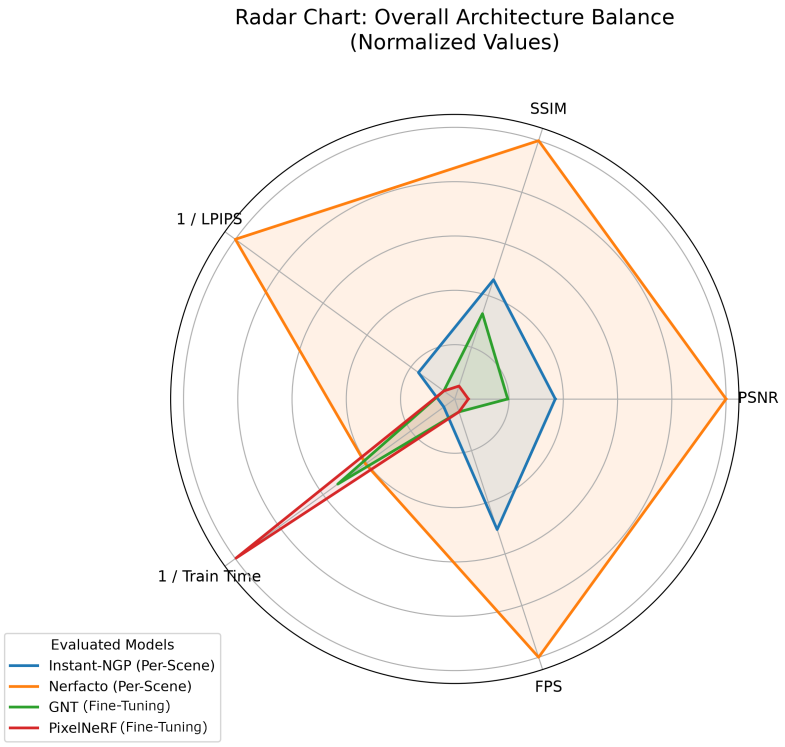


Figure 3. Multidimensional trade-offs under sparse-view constraints ($N = 10$). Metrics are max-normalized (outer perimeter = optimal), utilizing $1/$LPIPS and $1/$Train Time for uniform directionality. Per-scene models dominate fidelity and inference speed, whereas generalizable methods strictly excel in optimization efficiency.

Thus, the selection between regimes is not absolute, but entirely dictated by whether the deployment bottleneck lies in camera acquisition physics or rendering hardware capacity.

Table 5. Computational Performance and Memory on the fern scene.

Model	Regime	Time (s)	Inference (FPS)	VRAM Peak (GB)
PixelNeRF	ZS	39.02	0.003	1.60
	FT	496.82	0.006	1.92
GNT	ZS	**8.77**	0.007	1.97
	FT	392.15	0.008	1.97
Instant-NGP	N=10	1707.91	0.183	**1.09**
Nerfacto	N=10	423.45	**0.371**	1.10

4.4. Qualitative Analysis

The qualitative assessment visually corroborates the quantitative metrics, specifically illustrating the boundaries of each architectural paradigm. For visual clarity, novel views matching the exact poses from the test set were synthesized using perspective projection.

Figure 4 highlights the disparities in reconstruction quality on the highly detailed fern scene. As visually evident, neither PixelNeRF nor GNT achieved proper conver-

gence. PixelNeRF produced renderings with intense isotropic blurring, a typical artifact indicating the network's failure to accurately localize 2D projected features across extremely sparse views. GNT, on the other hand, demonstrated a complete generalization failure regarding the in-the-wild characteristics of the scene. This resulted in a representational visual collapse, heavily distorting the volumetric geometry into noise, which persisted even after the 10-view TTA.

Conversely, the per-scene architectures successfully captured the intricate foliage geometry. Nerfacto achieved the highest visual fidelity across all scenes (PSNR 27.11 dB, SSIM 0.875, LPIPS 0.171), recovering fine-grained details consistent with the ground truth resolution (Figure 4, bottom right).

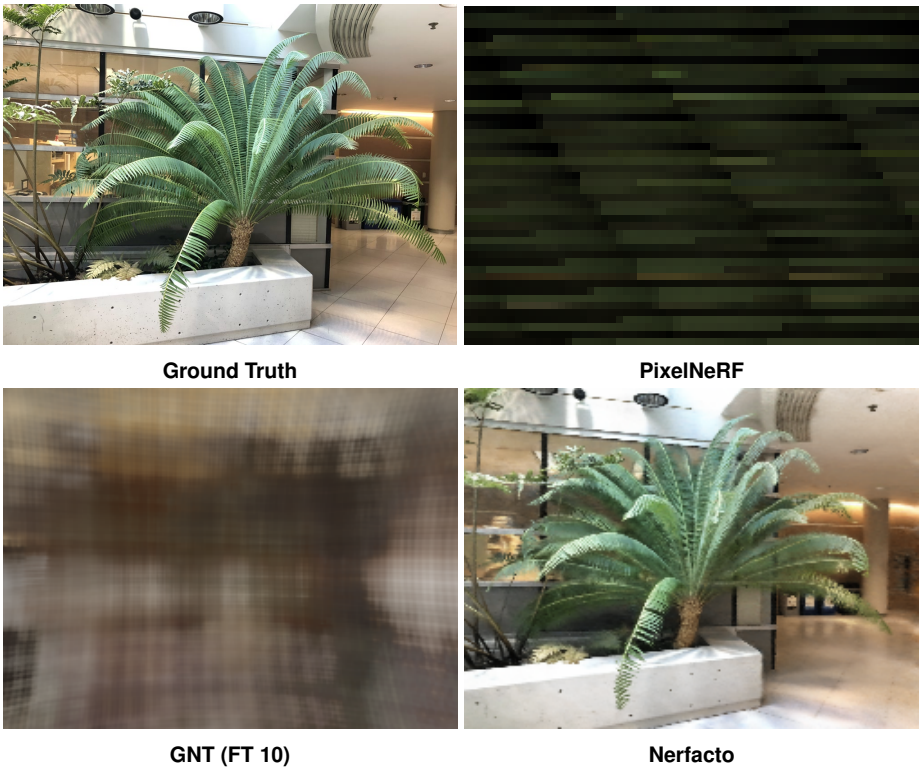


Figure 4. Qualitative comparison on the fern scene. Generalizable architectures (PixelNeRF and GNT) failed to converge under sparse constraints, resulting in severe blurring and visual collapse. In contrast, per-scene optimization models (specifically Nerfacto) successfully reconstructed the complex geometry.

5. Conclusion

This work developed a methodology for systematically characterizing the trade-offs between computational performance and reconstruction fidelity of Generalizable and Per-Scene NeRF architectures under sparse-view conditions. The experimental results demonstrate that Per-Scene models achieve substantially lower inference latency and VRAM consumption than generalizable models, making them well suited for deployment in resource-constrained environments. However, these benefits rely on dense multi-view supervision, restricting their usage in sparse-view acquisition scenarios, where generalizable architectures remain a practical alternative despite reduced reconstruction fidelity.

Although the proposed strategy provides a standardized and reproducible framework, some limitations should be acknowledged. The restricted hardware configuration of the experimental environment prevented the inclusion of architectures that rely on computationally intensive 3D cost volumes or strong geometric regularization, such as Reg-NeRF. Furthermore, the experimental validation was conducted on the LLFF dataset, and the adopted sparse-view sampling strategy exploits the sequential ordering of its image indices, which may limit its direct applicability to unordered image collections.

In this context, future work will focus on extending the proposed methodology to unbounded 360-degree datasets and more diverse acquisition scenarios, enabling a broader characterization of these trade-offs. This includes incorporating sparse-view 3D Gaussian Splatting variants [Seo et al. 2026, Song et al. 2025] into the benchmarking protocol, to assess whether their explicit representation offers a more favorable fidelity–efficiency trade-off than the implicit architectures evaluated here. Also, diffusion priors will be investigated as regularization mechanisms during FT to improve reconstruction quality.

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References

Adobe (2025). *Substance 3D Viewer (beta)*. Computer Software.

Hedman, P., Srinivasan, P. P., Mildenhall, B., Barron, J. T., and Debevec, P. (2021). Baking neural radiance fields for real-time view synthesis.

Heo, H., Kim, T., Lee, J., Lee, J., Kim, S., Kim, H., and Kim, J.-H. (2023). Robust camera pose refinement for multi-resolution hash encoding.

Kerbl, B., Kopanas, G., Leimkühler, T., and Drettakis, G. (2023). 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4).

Kulhanek, J. and Sattler, T. (2025). Nerfbaselines: Consistent and reproducible evaluation of novel view synthesis methods.

Laurino, A., Franceschini, A., Pesce, L., Cinci, L., Montalbano, A., Mazzamuto, G., Sancataldo, G., Nesi, G., Costantini, I., Silvestri, L., and Pavone, F. S. (2023). A guide to perform 3d histology of biological tissues with fluorescence microscopy. *International Journal of Molecular Sciences*, 24(7).

Mildenhall, B., Srinivasan, P. P., Ortiz-Cayon, R., Kalantari, N. K., Ramamoorthi, R., Ng, R., and Kar, A. (2019). Local light field fusion: Practical view synthesis with prescriptive sampling guidelines.

Mildenhall, B., Srinivasan, P. P., Tancik, M., Barron, J. T., Ramamoorthi, R., and Ng, R. (2020). Nerf: Representing scenes as neural radiance fields for view synthesis.

Müller, T. (2021). tiny-cuda-nn. https://github.com/NVlabs/tiny-cuda-nn. Version 1.7, BSD-3-Clause License.

Müller, T., Evans, A., Schied, C., and Keller, A. (2022). Instant neural graphics primitives with a multiresolution hash encoding. *ACM Transactions on Graphics*, 41(4):1–15.

Niemeyer, M., Barron, J. T., Mildenhall, B., Sajjadi, M. S., Geiger, A., and Radwan, N. (2022). Regnerf: Regularizing neural radiance fields for view synthesis from sparse inputs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5480–5490.

Schwarz, K., Liao, Y., Niemeyer, M., and Geiger, A. (2021). Graf: Generative radiance fields for 3d-aware image synthesis.

Seo, K., Hyun, S., Lee, M., and Heo, J.-P. (2026). Improving sparse-view 3dgs generalization via flat minima optimization.

Song, M., Lin, X., Zhang, D., Li, H., Li, X., Du, B., and Qi, L. (2025). D^2gs: Depth-and-density guided gaussian splatting for stable and accurate sparse-view reconstruction.

Tancik, M., Weber, E., Ng, E., Li, R., Yi, B., Wang, T., Kristoffersen, A., Austin, J., Salahi, K., Ahuja, A., Mcallister, D., Kerr, J., and Kanazawa, A. (2023). Nerfstudio: A modular framework for neural radiance field development. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Proceedings*, SIGGRAPH '23, page 1–12. ACM.

Wang, Z., Bovik, A. C., Sheikh, H. R., and Simoncelli, E. P. (2004). Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing*, 13(4):600–612.

Wang, Z., Varma, M., Wang, P., Chen, X., Chen, T., and Venugopalan, S. (2023). Is attention all that nerf needs?

Wu, H., Xiao, B., Codella, N., Liu, M., Dai, X., Yuan, L., and Zhang, L. (2021). Cvt: Introducing convolutions to vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 22–31.

Yu, A., Ye, V., Tancik, M., and Kanazawa, A. (2021). pixelnerf: Neural radiance fields from one or few images.

Zhang, R., Isola, P., Efros, A. A., Shechtman, E., and Wang, O. (2018). The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 586–595.